Research Approach v3.2: what changed from v3.1

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Version 3.2 replaces Task 5 (the framework) with Framework v2 and updates the sections that state our contribution. The reason is the related-work check of 9–11 Oct: two of our earlier novelty claims are already done by Madugalla et al. and NeuroAdaptX, so the paper now leads with what remains ours, run-time fidelity invariants that keep uncertainty. Everything else in v3.1 is unchanged.

The changes

Eight changes, in the order they appear in the document.

#	Section	v3.1 (7 Oct)	v3.2 (12 Oct)	Why
1	Positioning: gap statement	No work turns requirements into explanation requirements, "specifies them as user-chosen profiles", and checks fidelity	Same, but the new part is that fidelity is checked at run time on every adapted explanation, including AI wording, with uncertainty as one of five facts	Madugalla et al. already let users choose their own settings (p. 11)
2	Positioning: NeuroAdaptX contrast	Lists "requirements traced to software capabilities" as a difference	Difference is the run-time check with uncertainty; NeuroAdaptX checks once, by hand, before deployment, without uncertainty (p. 215)	NeuroAdaptX also traces its 10 design requirements to features

#	Section	v3.1 (7 Oct)	v3.2 (12 Oct)	Why
3	Positioning table	NeuroAdaptX: "no requirement-to-capability mapping"	"Traced to features, but not to data capabilities"; Madugalla row notes users choose settings	Correct both rows against the papers
4	Task 5: contribution	"This is the SE contribution" said of the capability table	Two contributions, invariants first, capability dependencies second; prior work named for what is not new	Lead with the strongest novel claim
5	Task 5: figure	One invariants check before the output	Check after every layer, fallback L4 → L2 → L1; data model types named	Shows exactly when the check runs and what happens on a fail
6	Task 5: invariants table	Invariant, example, grounding	Adds "How it is checked" against the `Forecast` fields	Makes each invariant testable in code
7	Task 5: capability table	Need → capability	Adds guidelines, data model field and status; two rows marked placeholder (landmark levels, action wording beyond "Possible")	Every row checkable; placeholders stated honestly
8	Task 5: layers	L3 alert mode: "every profile collapses to action only (G124)"	L3: one short line with all five facts, e.g. "Flooding possible here on Thursday. Get your go-bag ready today."	Action-only dropped place, time and likelihood, breaking the invariants rule; to confirm with Madam

Also updated: the summary now lists these changes, the worked trace names the data model functions, and the open questions list has three new questions (below).

Unchanged, and what we ask Madam

Unchanged: the research question, scope, Tasks 1–4b (guideline triage, prediction type and Hanwella scenario, XAI comparison, profiles, prototype, walkthrough), the paper plan, the rules and the risks. The fidelity invariants themselves (hazard, place, time window, likelihood, action) are the same five as in v3.1.

Three decisions for the 15 Oct meeting, explained with examples in `Supervisor_Questions_15Oct.pdf`:

- [ ] **Q1.** Lead with the run-time invariants and keep capability dependencies supporting, or present both equally?
- [ ] **Q2.** Alert mode keeps all five facts in one short line, or only hazard, place and action, as G124 suggests?
- [ ] **Q3.** Test the AI wording check on 10–20 prepared rewordings instead of a live model? And should it require the exact CAP word ("possible")?

The full v3.2 is `Research_Approach_v3.2_12Oct.pdf`; v3.1 stays as `Research_Approach_v3.1_7Oct.pdf` for comparison.